Joint Code–Channel Ambiguity in SP-AFDM

Method-design continuation from V2: residual correction, Bayesian hypothesis search, and their limits

Research manuscript and reproducible experimental record

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1. Research verdict and scope

The selected algorithmic candidate is a bounded, channel-marginalized, two-hypothesis codeword search. It addresses uncertainty in the channel used to generate and score the codeword list, rather than attempting to certify a truncated posterior under one potentially inaccurate channel estimate. The selected scientific line is **consumer-specific treatment of joint code–channel ambiguity**. This continues V2's channel-estimation concern, but does not assume that global posterior fidelity is necessary or that a missing-mass certificate is the only useful approach.

Four constructions were implemented: deterministic-list plus sampled-complement channel estimation; analytically channel-marginalized bounded hypothesis search; constrained pilot-phase design using a codeword-ambiguity surrogate; and optimizer-preserving channel-estimation coresets. The first method did not beat plain ordered-statistics decoding (OSD) at equal likelihood-evaluation budgets. The second produces the strongest receiver-level evidence, but regularized profile scoring and multistart EM are strong competitors. The transmitter-side surrogate does not yield robust empirical gains. The coreset establishes an important correction to the research target, but requires already acquired information and is not a cheap posterior-acquisition method.

No tested method has yet met the combined standard of defensible algorithmic novelty, favorable complete cost, and a confirmed advantage over the project's strongest receiver. The report supplies a finished conditional method section, executed comparative evidence, and specific failure mechanisms. It does not present classical importance sampling, Gaussian marginalization, mixture inference, OSD, or Carathéodory compression as inventions.

A particularly important qualification is that the main receiver experiments use **known path geometry**, including fractional Dopplers, and estimate unknown complex gains. This deliberately favorable oracle-geometry experiment isolates joint data/gain ambiguity. It is not an implemented solution to unknown delay/Doppler support acquisition. The derivation permits a declared integer dictionary, but a good prior on that dictionary cannot be assumed for free.

1.1 Continuity with V2 and the new runtime

V2 identified that list moments can be useful while a worst-case outside-mass certificate remains too conservative [P1, Sections 6–8]. It also showed that preserving MAP decisions need not preserve channel-estimation moments. Both conclusions remain valid. This continuation adds the converse qualification: a substantially different distribution can produce exactly the same particular channel update.

The V3 upload supplies an executable package closure. A one-frame run of its supplied smoke script at 7

dB completes, with zero decoded information errors, 238 raw errors, and approximately 5.12 seconds of receiver time. This checks availability, not a performance distribution. The smoke's `a0/direct40` settings are not silently relabeled as the separately configured formal I1/CodeFeedback receiver. The frozen files were not modified. No production-scale FER sweep was performed.

The existing full fractional-Doppler `physical_channel_matrix` function and coding infrastructure are reused. The independent physical construction and sign conversion documented in V2 remain the reference. This pass focuses on method development rather than repeating the entire historical audit. All new physical experiments retain full fractional propagation; no receiver-tail truncation is substituted into the propagation generator.

2. Model, targets, and information contracts

Let b be the information message, $c = Gb \pmod 2$ a valid binary codeword, and $d(c) \in \mathbb{C}^N$ its QPSK data vector. The received block is

$$y = H(g)(p + d) + w, \qquad H(g) = \sum_{i=1}^{S} g_i U_i, \qquad w \sim \mathcal{CN}(0, \sigma^2 I_N). \tag{1}$$

The pilot and data already include power scaling. The complex noise convention is $\mathbb{E}|w_n|^2 = \sigma^2$. In the experiments $|p_n|^2 = \rho$, $|d_n|^2 = P_d = 1 - \rho$, and each QPSK real component has magnitude $\sqrt{P_d/2}$. A particular block's pilot–data cross term need not vanish. No channel or truncated model is renormalized after observation generation.

The sampled physical assumptions are integer delays, adequate chirp-periodic prefix, full fractional Doppler, constant path parameters within a block, ideal synchronization, and proper sampled white noise. The handoff uses positive chirps and Doppler modulation with the negative-exponent unitary DFT. The corresponding $U_i = AC_{l_i}\Delta_{\nu_i}^{+}\Pi_{l_i}A^H$ are unitary individually. Their multipath sum need not be unitary.

The operational target is message error under complete computational cost. Three auxiliary objects are distinguished: a frozen plug-in data posterior for a chosen channel; a working joint distribution over data and gains; and the optimizer of a specified frozen-distribution CE objective. Exact statements about one do not automatically transfer to the others.

For the main marginalized model, assume a genuine exogenous prior

$$g \sim \mathcal{CN}(0, C_g), \qquad C_g \succ 0, \tag{2}$$

independent of the message and thermal noise. A pilot-based estimate is used to propose words, **not reused as if it were an independent prior**. A prior centered at an estimate from this same y would require a separate statistical interpretation. The initial matched experiment samples (2); the skeptical experiment intentionally normalizes the gains and therefore violates this Gaussian working prior.

The codebook is uniform in all new tests. Physical delays and Dopplers are supplied in the main method experiments. If the operators instead come from estimated integer support, the formulas define a working receiver model conditional on that support, not an exact posterior for the original unconditioned physical problem.

3. What CE actually needs: an estimating-equation identity

For each codeword define

$$\Phi_d = [U_1(p+d), \dots, U_S(p+d)], \quad K_d = \Phi_d^H \Phi_d, \quad b_d = \Phi_d^H y. \tag{3}$$

For a frozen distribution q, a fixed center g_0, and $\lambda > 0$, consider

$$J_q(g) = \mathbb{E}_q \|y - \Phi_d g\|^2 + \lambda \|g - g_0\|^2, \quad A_q = \mathbb{E}_q K_d + \lambda I, \quad g_q = A_q^{-1}(\mathbb{E}_q b_d + \lambda g_0). \tag{4}$$

This is V2's proper complex squared-residual objective. Its second-order expansion contains $\Gamma_{ij} = \mathrm{tr}(U_i^H U_j V_q)$, but the optimizer depends on the entire pair $(\mathbb{E}K_d, \mathbb{E}b_d)$, not Γ alone.

Proposition 1 (consumer-specific error identity). Define

$$r_d(g) = K_d g - b_d + \lambda(g - g_0).$$

For any two frozen distributions P and q on the same finite codebook,

$$\boxed{g_q - g_P = A_P^{-1} \mathbb{E}_P r_d(g_q).} \tag{5}$$

Furthermore,

$$\boxed{J_P(g_q) - J_P(g_P) = \bar{r}^H A_P^{-1} \bar{r}, \quad \bar{r} = \mathbb{E}_P r_d(g_q),} \qquad \|g_q - g_P\| \le \|\bar{r}\|/\lambda. \tag{6}$$

Proof. The normal equation under P is $A_P g_P = \mathbb{E}_P b_d + \lambda g_0$. Therefore $\mathbb{E}_P r_d(g_q) = A_P(g_q - g_P)$. Completing the square in the strictly convex quadratic J_P gives (6). The norm bound follows from $A_P \succeq \lambda I$. □

Unlike a total-variation bound, (5) allows large probability errors in directions that cancel in the estimating equation. It does not supply $\bar{r}$ cheaply: obtaining that expectation remains an inference task. It does identify the relevant task more narrowly than reproducing the entire posterior.

3.1 A distribution can be wrong while its CE optimizer is exact

Proposition 2 (optimizer-preserving finite coreset). For a finite codebook and fixed P, y, U, p, λ, g_0, a distribution q supported on at most $2S+1$ codewords exists such that $g_q = g_P$.

Proof. Stack the real and imaginary parts of $r_d(g_P)$ into a vector in $\mathbb{R}^{2S}$. Its P-weighted mean is zero. Hence zero lies in the convex hull of these vectors. Carathéodory's theorem gives a convex representation using at most $2S+1$ vectors. Those nonnegative weights sum to one and satisfy the same zero estimating equation. Since $\lambda > 0$, the corresponding quadratic has a unique optimizer, equal to g_P. Equivalently, one can repeatedly eliminate an affine dependence among more than $2S+1$ active vectors while preserving their convex combination. □

This result concerns one optimizer at one fixed state. It does not preserve every quadratic loss value, all future proximal centers, the posterior, LLRs, or the MAP decision. It is weaker than a coreset that preserves the complete least-squares objective. Exact least-squares coreset constructions are established prior art [R3].

Executed nondegenerate witness. For $N = 12$, three full fractional paths, $\rho = 0.25$, and $\sigma^2 = 4$, an exact 256-word posterior yields a nonzero optimizer with norm 1.01024. A linear-programming construction retains seven words. Its total variation from the original posterior is 0.577896, but the CE optimizer changes by only 1.78×10^{-16}. Retaining the seven highest-probability words and renormalizing instead changes the optimizer by 0.367712. An independent augmented least-squares implementation reproduces the equality.

The method requires the original optimizer or equivalent acquired statistics. It therefore **cannot be promoted as a solution to posterior acquisition**. Its value here is to demonstrate that global posterior fidelity is sufficient, not necessary, for this consumer.

4. First implemented candidate: deterministic head with sampled complement

V2's list truncation discards an unknown tail. A direct alternative is to integrate a short deterministic head exactly and estimate the complement, without first requiring a tight worst-case mass certificate.

Freeze a plug-in channel and unnormalized nonnegative code weights $f_d = \pi_d \exp[-\ell_d/\sigma^2]$. Let $\mathcal{L}$ be a fixed head, and draw $T_1, \dots, T_M$ independently from a proposal Q_T with positive probability everywhere in its complement. Form

$$\widehat{B} = \sum_{d \in \mathcal{L}} f_d (K_d + \lambda I) + \frac{1}{M} \sum_{m=1}^{M} \frac{f_{T_m}}{Q_T(T_m)} (K_{T_m} + \lambda I), \tag{7}$$

$$\widehat{a} = \sum_{d \in \mathcal{L}} f_d (b_d + \lambda g_0) + \frac{1}{M} \sum_{m=1}^{M} \frac{f_{T_m}}{Q_T(T_m)} (b_{T_m} + \lambda g_0), \qquad \widehat{g} = \widehat{B}^{-1} \widehat{a}. \tag{8}$$

The regularization is multiplied by the same estimated normalization as the data terms. Adding an unscaled regularizer to unnormalized data statistics would change the intended objective.

Proposition 3 (consistency, not finite-sample unbiasedness). On a finite codebook, with fixed positive-support Q_T and nonzero total weight, the sums in (7)–(8) are unbiased estimates of their complete unnormalized counterparts and converge almost surely as $M \to \infty$. Their solution converges to g_P. The solution itself need not be unbiased at finite M and need not improve marginal likelihood or FER.

Proof. The complement expectation of $f_T F(T)/Q_T(T)$ equals the omitted sum for any finite feature F. The strong law gives convergence. Positive weighted λI ensures invertibility; continuity of inversion at the limiting positive-definite matrix proves convergence of $\widehat{g}$. □

The implementation rescales all sampled and head weights by a common numerical maximum. The solver is invariant to that common scale. Unbiasedness is asserted for the underlying fixed-scale sums, not the randomly rescaled numbers saved in an implementation.

The proposal uses a most-reliable independent bit basis and a clipped product distribution on those information coordinates. A prefix-tree mass calculation excludes the deterministic head exactly, without rejection sampling. A 4,096-draw independent finite-code check confirms support exclusion and reported conditional densities; maximum empirical probability error 0.00741 is below the simultaneous 99% Hoeffding threshold 0.03398. This rules out a basic proposal-density bug as the explanation for the negative screen.

4.1 Discriminative result and pivot

The screen uses 128 paired configuration/observation instances, $N = 8, 12$, two phase seeds, two pilot powers and two noise levels. There are 32 distinct exogenous blocks reused across the four radio settings. It uses the actual channel to isolate approximation of a fixed posterior, not to claim an operational unknown-channel gain.

Total full-likelihood evaluations	Plain OSD: mean squared CE difference	Head plus sampled tail: mean squared CE difference
8	7.11×10^{-4}	1.31×10^{-3}
16	5.2×10^{-5}	1.019×10^{-3}

The reference is the exact frozen-posterior refit. Both arms receive the same observation, channel, support, and scoring rule. The larger nominal budget-32 control is retained in the raw data, but is not called exactly budget-matched because order-two OSD on the smaller code can generate only 22 words.

Decision. Generic complement sampling is not selected. The estimator is consistent, but finite-budget importance weights are poorly matched to the coupled likelihood. The result does not disprove residual-directed quadrature. It rejects the tested generic completion policy as a superior alternative to simple OSD. Monte Carlo joint channel estimation/detection also has substantial prior literature [R8].

This failure motivates a different mechanism: the dominant ambiguity may involve the channel used to propose and score words, not only the probability mass omitted under a fixed channel.

5. Proposed method: bounded channel-marginalized hypothesis search

5.1 Exact marginal score and conditional gain distribution

Under (1)–(2), set

$$B_d = C_g^{-1} + K_d/\sigma^2, \qquad \beta_d = b_d/\sigma^2, \qquad m_d = B_d^{-1}\beta_d, \qquad V_d = B_d^{-1}. \tag{9}$$

Proposition 4 (code-conditioned Gaussian integration). Conditional on a codeword,

$$g \mid d, y \sim \mathcal{CN}(m_d, V_d), \tag{10}$$

and the codeword's marginal log score, up to constants common to all words, is

$$\boxed{s_{\mathrm{marg}}(d) = \beta_d^H B_d^{-1}\beta_d - \log\det B_d + \log \pi_d.} \tag{11}$$

Proof. Expand the joint negative log density:

$$\frac{\|y - \Phi_d g\|^2}{\sigma^2} + g^H C_g^{-1} g = \frac{\|y\|^2}{\sigma^2} - \beta_d^H B_d^{-1}\beta_d + (g - m_d)^H B_d (g - m_d).$$

The last term is a proper complex Gaussian precision quadratic. Its integral is proportional to $\det(B_d)^{-1}$. Normalization of the prior and noise adds only codeword-independent constants, giving (10)–(11). □

A separate equivalent expression uses

$$C_d = \sigma^2 I + \Phi_d C_g \Phi_d^H, \qquad s_{\mathrm{marg}}(d) = -y^H C_d^{-1} y - \log\det C_d + \log \pi_d. \tag{12}$$

Equation (9) reduces the per-word factorization from N to S. Independent direct $N \times N$ covariance calculations are used for verification, not as the proposed online kernel.

Regularized joint-MAP **profiling** uses

$$s_{\mathrm{prof}}(d) = \beta_d^H B_d^{-1} \beta_d + \log \pi_d. \tag{13}$$

It omits the determinant, because maximizing over g does not integrate the volume of plausible gains. This makes profiling a particularly strong, same-list comparator: its main matrix computations are shared with marginalization.

This is classical Gaussian Bayesian integration. Joint channel uncertainty and Gaussian-mixture reduction were explicitly studied in MIMO receivers long before AFDM [R2]. The proposed bounded search is a concrete adaptation whose additional value must be measured, not assumed from (11).

5.2 Preserve distinct channel hypotheses before averaging

For normalized marginal weights w_d on a retained list, the list-conditioned gain mean and covariance are

$$\bar{g}_L = \sum_{d \in \mathcal{L}} w_d m_d, \quad V_{g,L} = \sum_{d \in \mathcal{L}} w_d [V_d + (m_d - \bar{g}_L)(m_d - \bar{g}_L)^H]. \tag{14}$$

Both within-word channel uncertainty and between-word ambiguity remain. Replacing all (d, m_d) pairs by $\bar{g}_L$ too early can obscure distinct channel explanations. Equation (14) does not assume the list contains all posterior mass. If it does not, it is explicitly a conditional approximation.

The operational search uses a few **individual conditional means** to propose additional codewords. It does not require a global outside-mass bound before doing useful computation.

5.3 Executed bounded algorithm

For a total budget $B = 16$, the implementation is:

1. Obtain a pilot-only initial gain $g^{(0)}$. It is used as a search seed, not as an independent prior. The executed initializer uses $\Phi_p = [U_i p]$ and penalty $(\sigma^2 + P_d \operatorname{tr} C_g) C_g^{-1}$.
2. Compute LMMSE-based bit reliabilities under $H(g^{(0)})$ and generate eight OSD candidates. Score them using (9)–(11).
3. Select the two highest marginal-score words from this initial set. Under each corresponding $H(m_d)$, generate a new OSD proposal list. Append unseen words, up to cumulative addition caps four and eight; unused capacity from the first branch can be used by the second. Never exceed 16 full likelihood evaluations.
4. Select the best marginal score on the union. Produce the list-conditioned gain mean (14). Preserve a profile-score output as a control. Do not report the largest normalized list weight as a certified correct-decoding probability.

The single-list control generates all 16 candidates under $g^{(0)}$. The branching method uses three OSD calls rather than one. Equal full-scoring budgets therefore do **not** imply equal total runtime. The branch code does not read true data, true gains, or a complete enumerated codebook. In the main tests it does receive oracle path geometry, which is separately declared.

A terminal-snapshot driver accepts only declared receiver inputs $y, p, U, C_g, G, g^{(0)}, \sigma^2, P_d$ and a systematic information-coordinate map. It validates dimensions and prior definiteness, returns candidate words and scores, and uses arbitrary-precision message identifiers for long codes. It does not install itself into I1 or replace a frozen feedback schedule.

5.4 What can genuinely be guaranteed

For fixed U, C_g, y and unchanged scoring, adding candidates to a retained union cannot decrease the largest retained marginal score. This is a set-inclusion statement, not a monotonic FER or gain-MSE theorem. A finite budget gives no guarantee of finding the globally best codeword. Exhaustive candidate coverage recovers exact marginal MAP under the stated model.

If the codebook and prior are complete and correct, the full posterior mean is Bayes-optimal for squared gain error. The list mean is optimal only for the explicitly conditioned list model. A wrong Gaussian prior, wrong path support, same-observation prior reuse, or incomplete candidate list breaks the stronger interpretation. The method does not claim convergence of an unspecified outer turbo iteration.

5.5 Degeneracies and failure conditions

For one unitary path, $p = 0$, and constant-modulus data, $K_d = \|d\|^2$ is constant across words. Thus the determinant term is constant, and profiling and marginalization have identical rankings. SP changes $\|p + d\|^2$ from word to word, so the determinant can become relevant, but relevance does not imply a large FER advantage.

Unregularized profiling can overfit if sufficiently many channel coordinates make every Φ_d full row rank: each word can then fit y exactly. Positive prior precision avoids a singular problem, but its modeling assumptions become central. Incorrect support remains incorrect support; a polished Gaussian integral cannot repair missing operators.

Proposition 5 (an exact integer representation can have an uninformative prior). For the complete integer AFDM operator basis with N^2 distinct delay–Doppler pairs,

$$\sum_{l,q} U(l,q) x x^H U(l,q)^H = N\|x\|^2 I. \tag{15}$$

If all integer coefficients are independent $\mathcal{CN}(0, 1/N^2)$, then

$$y \mid d \sim \mathcal{CN}\left(0, [\sigma^2 + \|p + d\|^2/N] I\right). \tag{16}$$

Proof. Move xx^H through the unitary DAFT. Summing the integer Doppler modulations cancels all off-diagonal entries. The prefix phases have unit modulus and do not alter the resulting diagonal. Summing all cyclic shifts gives $\|x\|^2$ on every diagonal entry, with the factor N from the Doppler sum. Gaussian independent coefficients then yield (16). □

Only scalar total energy remains in the integrated likelihood. With no pilot and constant-modulus data, every codeword has the same marginal likelihood. This is not an impossibility of integer representation: a coherent physical-path prior induces structured, generally correlated integer coefficients. Equation (16) diagnoses a bad indiscriminate prior on an exact but overcomplete parameterization. The independent verifier checks (15) at $N = 5$.

6. Complexity and relation to classical methods

For each candidate, forming Φ_d, K_d, and b_d costs the operator-application cost plus $O(NS^2)$. Factoring B_d costs $O(S^3)$, and triangular solves provide the conditional mean, log determinant, and any required covariance actions. The reference implementation stores dense physical U_i, so its application cost is $O(SN^2)$ per word. A monomial integer representation permits $O(SN)$ application, while known physical paths can exploit transforms and shifts. Those optimizations were not silently credited to the dense reference timing.

For B evaluated words, the principal score cost is

$$O\Big(B[C_{\text{apply}} + NS^2 + S^3]\Big)\,, \tag{17}$$

plus pilot initialization, LMMSE reliability generation, GF(2) OSD reprocessing, and up to three OSD proposal invocations. Candidate selection also requires deduplication. The main implementation returns in-list statistics without a missing-mass certificate. Comparing a 16-word method with exhaustive small-code EM by candidate count alone is inappropriate because their per-word work differs.

Exact EM is included as a strong reference under the same Gaussian gain prior. At iteration t, enumerate the exact plug-in data posterior for $g^{(t)}$ and update

$$g^{(t+1)} = \Big[\mathbb{E}_{q_t} K_d + \sigma^2 C_g^{-1}\Big]^{-1} \mathbb{E}_{q_t} b_d. \tag{18}$$

The penalized marginal log likelihood is nondecreasing under exact E/M steps. The new code checks this property and an independent augmented least-squares implementation reproduces the updates. EM remains a point-parameter optimizer and can converge to different basins; its monotonicity does not mean global message-risk optimality. A three-start, five-step EM control uses the pilot seed and two profile-conditioned seeds, with all selection costs identified.

7. Numerical evidence for the selected candidate

7.1 Matched-prior main experiment

The main experiment uses $N = 8, 12$, code dimensions $k = 6, 8$, one or three physical paths, $\rho \in \{0.15, 0.5\}$, and $\sigma^2 \in \{0.15, 0.5\}$. Each cell has 16 independently generated channel/message/noise realizations, giving **256 independent draws across the balanced grid**. Arms are paired on the same draws. The three-path coordinates are $(0, 0.41), (1, -0.27), (2, 1.12)$, and the gain covariance is proportional to $\mathrm{diag}(1, 0.65^2, 0.4^2)$ with unit trace. One-path gain variance is one. Actual gains are not normalized in this experiment.

Method	Failures / 256	Mean squared gain error
Pilot estimate, exact plug-in codeword decoding	133	0.353947
Exact five-step EM, one start	103	0.231176
Exact five-step EM, three starts	83	0.193903
Complete codebook, marginalized channel	73	0.112005
Complete codebook, profiled channel	74	0.150877
One OSD list, marginalized, 16 words	89	0.192276
Same OSD list, profiled, 16 words	88	0.224958
Two-hypothesis marginalized search, 16 words	88	0.182384
Actual channel, exact codeword decoding	44	0

Mean squared gain error is $\mathbb{E}\|\widehat{g} - g\|^2$, **not normalized channel NMSE**. The priors do not fix each realization's gain norm. Full-codebook results are oracle-computation references; geometry is common to all arms.

Against one-start EM, the branch method rescues 22 frames and harms seven. Against three-start EM, it rescues three and harms eight. On the same list, marginalization versus profiling rescues one and harms two. Thus the data do not support claiming that the determinant correction is a uniquely effective decoding mechanism. The marginalized list mean has lower average gain squared error than the profiled point output, but those are different loss-targeted estimators and do not imply an equivalent FER gain.

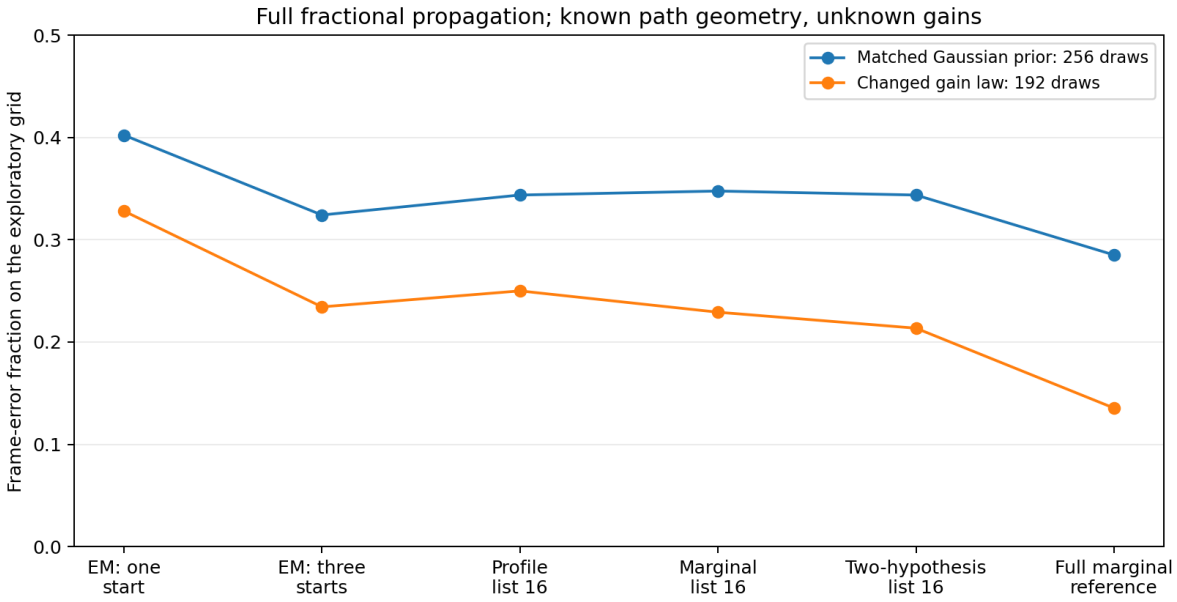


Figure 1: Paired exploratory error fractions under full fractional propagation. The skeptical set changes the true gain law while keeping the working Gaussian prior.

7.2 Skeptical pass: deliberate gain-prior mismatch

A separate stream with 12 frames per cell gives 192 new draws. Actual Gaussian gains are normalized to unit norm; the receiver retains its previous Gaussian working prior. Other grid settings are unchanged.

Method	Failures / 192	Mean squared gain error
Pilot estimate	76	0.385731
Five-step EM, one start	63	0.255801
Five-step EM, three starts	45	0.216840
Complete marginalized model	26	0.107684
Complete profile model	32	0.107824
One marginalized list, 16 words	44	0.209290
Same profiled list, 16 words	48	0.221238
Two-hypothesis search, 16 words	41	0.210457
Actual-channel exact decoding	19	0

The branch method rescues 28 and harms six relative to one-start EM. It rescues eight and harms four relative to three-start EM. These results make the candidate worth retaining, but do not establish population superiority over the strongest controls. The complete marginalized model is no longer the true Bayes reference under the changed gain law. Main and skeptical sets are not pooled as though generated by the same distribution.

7.3 Longer-code skeptical check: the list-acquisition barrier

To avoid drawing the conclusion only from unusually small codebooks, a separate nonenumerated check uses $N = 24, 48$, rate-near-half Gallager codes, three paths, $\sigma^2 = 0.3$, two pilot powers and 16 fresh frames per cell. A conventional five-round soft CE loop using fresh LMMSE likelihoods and a 20-iteration LDPC decoder is included. It is not the frozen I1 implementation and is not labeled as such.

N	ρ	Marginal list 16	Profile list 16	Two-hypothesis list 16	Soft CE, five rounds
24	0.15	11/16	11/16	10/16	12/16
24	0.50	12/16	12/16	13/16	14/16
48	0.15	13/16	13/16	9/16	7/16
48	0.50	13/16	13/16	10/16	11/16
Total		49/64	49/64	42/64	44/64

In these 64 draws, every marginal/profile/branch list failure coincides with the transmitted word being absent from that arm's candidate set. No retained true word is misranked in this particular check. This is empirical, not a general theorem. It identifies the immediate bottleneck: **candidate acquisition, not refinement of the score among the existing words**. A better outside-mass certificate would diagnose the issue but would not generate the missing word.

The list algorithms remain high-error methods in this difficult cold-start test. The result is not a production-code advantage. The branch method broadens useful hypotheses, but a conventional soft-CE loop is superior at one of the stronger $N = 48$ cells. This prevents an unsupported universal ordering.

7.4 Dedicated timing

The benchmark performs 31 repeated single-frame kernels after warm-up on a representative three-path fixture. It excludes shared initial-pilot estimation and static small-codebook creation, but includes online proposals, candidate response formation, scoring, and required refits. All raw timings are saved. These are microbenchmark medians, not confidence intervals or production costs.

Kernel	$N = 8$	$N = 12$
One marginalized OSD list, 16	850.06 μs	828.03 μs
Two-hypothesis marginalized search, 16	2106.79 μs	2349.10 μs
Complete marginalized small codebook	171.01 μs	751.46 μs
Complete five-step EM	2218.44 μs	3726.45 μs

The branch method is more expensive than one list because it generates two additional reliability/OSD proposals. It is faster than this unoptimized exact small-code EM kernel at the selected inputs, but the table cannot establish superiority to optimized BP/EM or I1. Complete marginalization is particularly cheap on the tiny codebooks and does not scale that way to long codes.

8. Broader alternative: ambiguity-aware pilot design

The receiver results motivate a transmitter-side test: could a fixed SP pilot distinguish code/channel explanations before inference becomes difficult? This is genuinely different from tuning the list or its certificate.

For the matched Gaussian gain model, the conditional observation is zero-mean proper Gaussian with covariance $C_d(p) = \sigma^2 I + \Phi_d(p) C_g \Phi_d(p)^H$. The pairwise Bhattacharyya coefficient is

$$\mathrm{BC}(d, e; p) = \frac{\sqrt{\det C_d(p) \det C_e(p)}}{\det[(C_d(p) + C_e(p))/2]}. \tag{19}$$

Proposition 6 (full-codebook risk surrogate). For M equiprobable codewords and the matched model,

$$P_e^{\mathrm{MAP}} \le \min\left\{1, \frac{1}{M} \sum_{d<e} \mathrm{BC}(d, e; p)\right\}. \tag{20}$$

Proof. At each observation, the sum of class densities minus their maximum is bounded by the sum of pairwise minima. Each minimum is at most the geometric mean. Integrating and dividing by M yields (20). Integrating two proper Gaussian square-root densities gives (19), using the determinant identity for their harmonic-average precision. □

The executed optimization samples up to 128 unequal pairs and minimizes their average coefficient. It is therefore a **sampled surrogate**, not the complete certified bound (20). Pilot phases vary with fixed DAFT-domain magnitude $\sqrt{\rho}$. Line search accepts only surrogate improvement while keeping the maximum time-domain PAPR over the full tiny codebook no larger than the original pilot's maximum. Twenty-four gradient attempts are used. A separate set of 64 random phase perturbations supplies a simple feasible competitor, selected by the same training surrogate.

Test	Original pilot	Optimized pilot	Best feasible random training candidate
$N = 8$, design geometry	25/96	26/96	23/96
$N = 8$, different fractional Dopplers	35/96	34/96	26/96
$N = 12$, design geometry	20/96	17/96	21/96
$N = 12$, different fractional Dopplers	24/96	25/96	23/96

The objective decreases from 0.139594 to 0.138409 at $N = 8$, and from 0.056119 to 0.052587 at $N = 12$. The latter improvement does not persist as a decoding advantage after changing Dopplers. All terminal decisions use complete channel-marginalized codebook decoding, so a weak iterative detector is not the explanation. The physical channel gains/noise are paired across pilots. The PAPR guarantee concerns these tiny codebooks, not all future codes.

Decision. This pilot candidate is not selected. The test rejects the tested sampled objective/phase search as a robust improvement, not all pilot design. Pairwise-error-based superimposed training design is itself not a new problem [R7]. Enlarging the search blindly would not constitute evidence of a new mechanism.

9. Independent verification and reproducibility

The independent verification script imports no candidate algorithm module. For every main and skeptical small-code observation it reconstructs candidate response matrices directly, evaluates (12) using $N \times N$ covariance inverses, and checks the $S \times S$ formulas modulo codeword-independent constants. It reconstructs complete and list decisions from saved candidate indices and checks the reported squared gain errors. Five-step EM is independently implemented using stacked, weighted, augmented least squares rather than the normal equations used in the candidate code. Coreset equality is also checked by this different solve path.

The verifier passes **3,588 comparisons**, with maximum absolute discrepancy 1.19×10^{-12}. The maximum EM gain discrepancy is 3.69×10^{-15}. The full integer-basis prior-collapse identity differs by 1.07×10^{-14}. These comparisons are numerical consistency checks, not independent research replications or an independent human review.

A fresh same-environment execution reproduces every one of the **2,368 arrays** in the main experiment and its CSV byte-for-byte. Elapsed timing is not part of that equality claim. The exact-complement sampler has its separate probability check. The authorized-input snapshot driver is exercised on a saved small fixture. The executed V3 smoke check is distinct from all these new methods.

Reproduce the principal study from the package root:

```
export OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1
python src/screen_tail.py
python src/run_collapsed.py --out collapsed_main
python src/run_collapsed.py --out collapsed_skeptical \
  --offset 777 --frames 12 --mismatch
python src/longer_check.py
python src/pilot_design.py
```

```
python src/coreset_check.py
python src/coreset_broad.py
python src/check_sampler.py
python src/independent_verify.py
python src/benchmark_methods.py
```

Runtime measurements will vary. The code is a research reference and retains its exact tested operations; it is not described as production-optimized. Explicit inverses occur for small prior/covariance reference operations. A production version should use cached factorizations, structured operators, and triangular solves, with new timing evidence rather than transferred percentages.

10. Novelty audit and manuscript claim boundaries

Public literature was checked through September 19, 2026. Primary source records and access levels are retained in `literature.json`. We distinguish directly inspected manuscripts from author/institution abstract records. None of their numerical curves was reproduced as a faithful external baseline in this pass.

Prior work	Established scope	Consequence here
Zheng et al., SP-AFDM [R4]	Pilot placement and iterative CE/detection	SP iteration is infrastructure, not novelty.
Huang et al., SP-OTFS [R5]	GCE-BEM and soft joint inference	Integer/BEM soft reception is established.
Feng et al., soft-output OSD/GRAND/GCD/SCL [R1]	Modeling unvisited-codeword probability under stated assumptions	Outside-list correction cannot be claimed as a new general idea.
Jia et al., joint MIMO mixture receiver [R2]	Channel-uncertainty-aware detection and Gaussian-mixture approximation	Marginalization and maintaining channel hypotheses belong to a classical family.
Maalouf et al., exact LMS coresets [R3]	Carathéodory-based exact least-squares compression	Optimizer-preserving subset construction is not a new inference family.
Luo et al., fractional AFDM-SP SBL/GAMP [R6]	Data-aided sparse inference with fractional parameters	Broad AFDM/SP/fractional/sparse combinations are occupied.
Yang, CSI dissertation [R7]	Superimposed-training design and pairwise-error objectives	A codeword-ambiguity pilot criterion is not novelty by itself.
Jing et al., sparse-channel MCMC reception [R8]	Joint stochastic channel/data inference	Monte Carlo completion is classical, even if a particular sampling interface differs.
Ranasinghe et al., PBiGaBP AFDM [R9]	Bayesian joint channel/data estimation	A scalable literature-supported Bayesian comparator remains necessary.

The available evidence does not establish that the precise two-hypothesis search has a historically unique implementation. More importantly, it does not yet demonstrate an advantage that a simpler profile-list or multistart method cannot obtain. The central proposed method is therefore presented as the strongest **tested candidate**, not as a proven new Transactions contribution.

The strongest theoretical correction is the distinction between posterior approximation and **consumer-specific estimating-equation approximation**. It sharpens the objective but does not, by itself, give an inexpensive acquisition algorithm. A publishable continuation would need to exploit a particular SP-AFDM structure to acquire useful code/channel hypotheses or residual statistics more efficiently than the classical controls. This obligation is not disguised as a theorem or delegated to an unrun oracle.

11. Claim-to-evidence map and remaining confirmation

Claim	Status	Evidence
CE error is controlled exactly by the estimating-equation residual	Proved	Proposition 1; coreset and augmented-solve checks
Global posterior fidelity is not necessary for one CE optimizer	Proved and witnessed	Proposition 2; `coreset_broad_raw.npz`
Generic head/tail completion improves same-budget OSD	Not supported	`tail_screen.csv`; sampler validation
Marginal score differs from profile by log determinant	Proved and independently checked	Proposition 4; `independent_verify.py`
Branching can improve the tested single-trajectory cold start	Exploratory empirical support	Main and skeptical paired tables
Branching dominates profile or multistart EM	Not supported	Same-list controls and three-start EM
Larger-code failures are list omissions in this check	Empirical scope only	`longer_screen.csv`, `longer_raw.npz`
IID full-integer coefficient prior can erase directional information	Proved and checked	Proposition 5; complete-dictionary verification
Sampled pilot objective provides a robust FER advantage	Not supported	`pilot_screen.csv`, shifted-Doppler check
Complete I1/production cost and FER superiority	Unresolved, not executed	V3 smoke is availability only; confirmation protocol

A terminal snapshot interface and a frozen confirmation configuration are delivered. They are executable on explicitly supplied receiver snapshots, but no full I1 integration or confirmation campaign is claimed. The V3 runtime now exists; the remaining work is **scientific integration and a matched comparison**, not an invented missing-dependency barrier.

The confirmation protocol separates selection from evaluation, freezes candidate budget and branch rules, includes profile rescoring, extra-decoder/iteration and multistart controls, and requires complete runtime accounting. Receiver-provided geometry and priors must be recorded as estimated versus oracle. The current favorable known-geometry results cannot be promoted to unknown-support operation without that comparison.

Final judgment. Continue the project around resolving joint code–channel ambiguity with statistics chosen for their actual CE consumer. Retain bounded channel-marginalized hypothesis search as the strongest concrete receiver candidate from this pass. Do not promote the sampled-tail correction, pilot surrogate, exact coreset, or marginalization formula as a completed new receiver contribution. The run supplies implemented methods, controlled positive and negative evidence, and a narrower theoretical target; it does not claim that a publication-level advantage has already been demonstrated.

References and source record

[P1] Supplied V2 report, *From MAP Preservation to Posterior-Moment Fidelity in Coded SP-AFDM*, September 19, 2026. Packaged unchanged in `prior_v2/spafdm_v2_study/docs/research_report.md`.

[P2] Supplied `spafdm_receiver_handoff_20260917-v3.zip`, including executable package closure and smoke adapter. Original archive SHA-256 is recorded in the environment file. New smoke output: `results/v3_smoke.json`.

[R1] J. Feng, K. R. Duffy, and M. Médard, *Leveraging Code Structure to Improve Soft Output for GRAND, GCD, OSD, and SCL*, arXiv:2503.16677v1, March 20, 2025. HTML inspected, including the independent-bit and unvisited-codeword assumptions.

[R2] Y. Jia, C. Andrieu, R. J. Piechocki, and M. Sandell, *Gaussian Approximation Based Mixture Reduction for Joint Channel Estimation and Detection in MIMO Systems*, IEEE Transactions on Wireless Communications, vol. 6, pp. 2384–2389, July 2007. DOI: 10.1109/TWC.2007.05911. Author-institution abstract and publication record inspected; detailed algorithm not reproduced.

[R3] A. Maalouf, I. Jubran, and D. Feldman, *Fast and Accurate Least-Mean-Squares Solvers*, arXiv:1906.04705, 2019. Author manuscript record and exact-coreset scope inspected.

[R4] K. Zheng, M. Wen, T. Mao, L. Xiao, and Z. Wang, *Channel Estimation for AFDM With Superimposed Pilots*, arXiv:2404.10232, 2024; IEEE Transactions on Vehicular Technology, 2025. Original pilot/iterative scope, not a claim that every case here is covered there.

[R5] F. Huang, Q. Guo, Y. Zhang, and Y. Zakharov, *Message Passing-Based Joint Channel Estimation and Signal Detection for OTFS with Superimposed Pilots*, arXiv:2309.08177v2; IEEE Transactions on Vehicular Technology, 2024. DOI: 10.1109/TVT.2024.3378833.

[R6] Y. Luo et al., *Data-aided Channel Estimation and Sensing With Sparse Bayesian Learning for AFDM-ISAC System*, arXiv:2607.18881v1, July 21, 2026. HTML inspected. Used to constrain novelty, not as a reproduced baseline.

[R7] J. Yang, *Channel State Information in Multiple Antenna Systems*, Georgia Institute of Technology dissertation, 2006. Institution's indexed abstract consulted for superimposed-training/pairwise-error design scope; direct page access returned 403 on recheck. No detailed theorem is attributed to inaccessible content.

[R8] L. Jing et al., *Joint Channel Estimation and Detection Using Markov Chain Monte Carlo Method over Sparse Underwater Acoustic Channels*, IET Communications, 2017. DOI: 10.1049/iet-com.2016.1339. Publisher/author-record scope checked; algorithm not reproduced.

[R9] K. R. R. Ranasinghe et al., *Joint Channel, Data, and Radar Parameter Estimation for AFDM Systems in Doubly-Dispersive Channels*, arXiv:2405.16945v4; IEEE Transactions on Wireless Communications, vol. 24, no. 2, 2025. DOI: 10.1109/TWC.2024.3510935. Primary record and Bayesian parametric scope inspected.